

Lecture - 20 - RL Debugging & Diagnostics

- 1.) Build a model / simulator of ~~helix~~ helicopter.
- 2.) Choose a reward function.

Say $R(s) = -\|s - s_{\text{desired}}\|^2$

(s = helicopter position)

- 3.) Run reinforcement ~~learning~~ learning (RL) algorithm to fly helicopter in simulation to maximize $E[R(s_0) + R(s_1) + \dots + R(s_T)]$, and get a learned policy π_{RL} .

Suppose you do this, and the resulting controller π_{RL} gives much worse performance than your human pilot. What to do next?

Improve model / simulator?
 Modify reward function R ?
 Modify RL algorithm?

The controller given by π_{RL} performs poorly.

Suppose that:

- 1.) The helicopter simulator is accurate.
- 2.) The RL algorithm correctly controls the helicopter (in simulation), so as to maximize expected pay off $V^{\pi_{RL}, s_0} = E[R(s_0) + R(s_1) + \dots + R(s_T)]$

3.) Maximizing expected pay off corresponds ~~to~~ to correct autonomous flight.

Then: the learned controller π_{RL} should fly well on the actual helicopter.

Diagnostics:

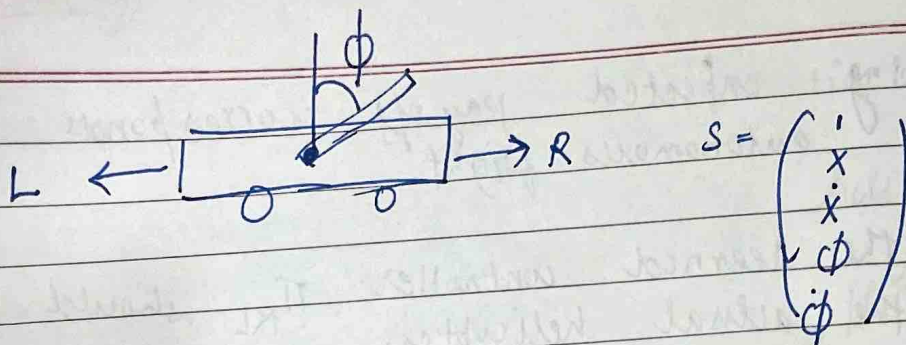
1.) If π_{RL} flies well in simulation, but not in real life, then the problem is in the simulator. Otherwise:

2.) Let π_{human} be the ~~best~~ policy ~~for~~ human control policy. If $V^{\pi_{RL}}(s_0) < V^{\pi_{human}}(s_0)$, then the problem is in the reinforcement learning algorithm. (Failing ~~to~~ to maximize the expected payoff.)

3.) Otherwise, then the problem is in the cost function. (maximizing it "doesn't" correspond to good autonomous flight).

Policy search ("direct policy search")

$$v^* \rightarrow \pi^*$$



$$y \approx h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$$a \approx \pi_{\theta}(s) = \frac{1}{1 + e^{-\theta^T s}}$$

New definition $\rightarrow$ A stochastic policy is a function $\pi: S \times A \rightarrow R$ when $\pi(s, a)$ is the probability of taking action a in state s .

$$\left(\sum_a \pi(s, a) = 1 \right)$$

$$\pi_{\theta}(s, "R") = \frac{1}{1 + e^{-\theta^T s}} = \frac{1}{1 + e^{-\phi}}$$

$$\pi_{\theta}(s, "L") = 1 - \frac{1}{1 + e^{-\theta^T s}} = 1 - \frac{1}{1 + e^{-\phi}}$$

$$s = \begin{pmatrix} 1 \\ x \\ \dot{x} \\ \phi \\ \dot{\phi} \end{pmatrix} \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{bmatrix}$$

Goal: Find θ so that when we execute $\pi_{\theta}(s, a)$ we maximize,

$$\max_{\theta} E \left[R(s_0, a_0) + R(s_1, a_1) + \dots + R(s_T, a_T) \mid \pi_{\theta} \right]$$

Note: Here, s_0 is a fixed initial state

$$\max_{\theta} E \left[R(s_0, a_0) + R(s_1, a_1) + \dots + R(s_T, a_T) \mid \pi_{\theta} \right]$$

taking $T=1$

$$V = \sum_{s_0, a_0, s_1, a_1} P(s_0) P(s_0, a_0, s_1, a_1) \cdot \underbrace{[R(s_0, a_0) + R(s_1, a_1)]}_{\text{Pay-off}}$$

$$= \sum_{s_0, a_0, s_1, a_1} P(s_0) \pi_{\theta}(s_0, a_0) \cdot P_{s_0, a_0}(s_1) \cdot \pi_{\theta}(s_1, a_1) \cdot [\text{payoff}]$$

Reinforce algorithm $\rightarrow$

Loop {

sample s_0, a_0, s_1, a_1

compute pay-off = $R(s_0) + R(s_1) + \dots$

update:

$$\theta = \theta + \alpha \left[\frac{\nabla_{\theta} \pi_{\theta}(s_0, a_0)}{\pi_{\theta}(s_0, a_0)} + \frac{\nabla_{\theta} \pi_{\theta}(s_1, a_1)}{\pi_{\theta}(s_1, a_1)} \right] \text{payoff}$$

}

$$\nabla_{\theta} \sum_{s_0 a_0 s_1 a_1} P(s_0) \pi_{\theta}(s_0 a_0) P_{s_0 a_0}(s_1) \pi_{\theta}(s_1 a_1) [\text{pay off}]$$

$$\sum P(s_0) \pi_{\theta}(s_0 a_0) \left(\frac{\nabla_{\theta} \pi_{\theta}(s_0 a_0)}{\pi_{\theta}(s_0 a_0)} P_{s_0 a_0}(s_1) \pi_{\theta}(s_1 a_1) + \right.$$

$$\left. P(s_0) \pi_{\theta}(s_0 a_0) P_{s_0 a_0}(s_1) \pi_{\theta}(s_1 a_1) \frac{\nabla_{\theta} \pi_{\theta}(s_1 a_1)}{\pi_{\theta}(s_1 a_1)} \right]$$

$$= P(s_0 a_0 s_1 a_1) \left[\frac{\nabla_{\theta} \pi_{\theta}(s_0 a_0)}{\pi_{\theta}(s_0 a_0)} + \frac{\nabla_{\theta} \pi_{\theta}(s_1 a_1)}{\pi_{\theta}(s_1 a_1)} \right] [\text{pay off}]$$